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ORIGINAL ARTICLE

Association Between 24-Hour Systolic Blood Pressure Time in Target Range and Mortality

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BACKGROUND: Time in target range (TTR) reflects the proportion of time blood pressure (BP) remains within a defined range, integrating BP variability and control. We examined associations of systolic BP (SBP) TTR during ambulatory BP monitoring with cardiovascular and all-cause mortalities.

METHODS: Patients from the Spanish Ambulatory BP Monitoring Registry who were receiving antihypertensive medications or who had sustained or masked hypertension without treatment, defined by office BP ≥140/90 mm Hg and 24-hour BP ≥130/80 mm Hg. TTR was estimated by linear interpolation between consecutive SBP recordings obtained from ambulatory BP monitoring and expressed as the proportion of time SBP remained within 120 to 134 mm Hg during daytime and 110 to 119 mm Hg during nighttime, from which 24-hour TTR was derived. Associations with mortality were assessed by Cox regression adjusted for demographic and clinical variables.

RESULTS: A total of 48 687 patients (46% women) were analyzed. Over a median follow-up of 9.7 years, 6502 deaths occurred, including 2185 cardiovascular deaths. Higher 24-hour TTR was associated with lower all-cause mortality (hazard ratio, 0.83 per 1-SD increment [95% CI, 0.80–0.85]). Similarly, higher 24-hour TTR was associated with lower cardiovascular mortality (hazard ratio, 0.80 per 1-SD increment [95% CI, 0.76–0.84]). Both associations remained significant after adjusting for the mean 24-hour SBP and SBP variability.

CONCLUSIONS: Higher 24-hour SBP TTR derived from ambulatory BP monitoring was independently associated with lower all-cause and cardiovascular mortalities. ***(Hypertension.* 2026;83:00–00. DOI:10.1161/HYPERTENSIONAHA.125.26112.)** • **Supplement Material**.

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Key Words: anticoagulants ■ blood pressure monitoring, ambulatory ■ hypertension prognosis ■ stroke

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Short-term blood pressure (BP) variability, as measured by 24-hour ambulatory BP monitoring (ABPM), is associated with an increased risk of cardiovascular outcomes, independent of the mean 24-hour BP level.[1–3] An emerging metric in assessing BP variability and control is time in target range (TTR), which has gained increasing attention in hypertension management. The concept of TTR originated in the fields of anticoagulation and diabetes care, where it is used both as a clinical target and outcome measure.[4,5] More recently, TTR has also been applied to hypertension.[6,7] TTR quantifies the proportion of time that systolic BP (SBP) remains within a predefined target range over multiple time points.[8] Unlike mean BP, TTR integrates information on BP levels and variability. For example, a patient may have a mean BP within the target range but exhibit high BP variability, resulting in fewer individual readings within the desired range. Because both elevated BP levels and increased BP variability are associated with all-cause and cardiovascular mortalities,[9,10] TTR may offer additional prognostic value. In a post hoc analysis of the SPRINT (SBP Intervention Trial), comparing intensive (unattended office SBP <120 mm Hg) versus standard (unattended office SBP <140 mm Hg) BP targets in patients with hypertension and increased cardiovascular risk, TTR calculated from 3 office BP recordings taken over the initial 3 months was associated with major adverse cardiovascular events, even after adjusting

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NOVELTY AND RELEVANCE

What Is New?

Spending more time within the target range during ambulatory blood pressure monitoring was associated with lower risks of all-cause and cardiovascular deaths, independent of mean blood pressure and its variability.
This benefit was observed in patients with elevated or well-controlled blood pressure, but not in those with low levels.
The association was seen across most hypertension phenotypes, except white-coat uncontrolled hypertension.

What Is Relevant?

Time in target range provides additional prognostic information beyond mean blood pressure and variability for mortality risk assessment.

Clinical/Pathophysiological Implications?

Future studies should evaluate whether interventions that increase time in the target range lead to improved patient outcomes.

Nonstandard Abbreviations and Acronyms

ABPM	ambulatory blood pressure monitoring
ACCORD-BP	Action to Control Cardiovascular Risk in Diabetes-Blood Pressure
BP	blood pressure
DBP	diastolic blood pressure
HR	hazard ratio
SBP	systolic blood pressure
SPRINT	Systolic Blood Pressure Intervention Trial
TTR	time in target range

for mean SBP or BP variability.[6] The target SBP ranges were 110 to 130 and 120 to 140 mm Hg in the intensive and standard treatment arms, respectively.[6]

When calculating TTR from home or office BP measurements using linear interpolation, this approach may inadequately capture rapid or nonlinear fluctuations and is more susceptible to the influence of outliers due to the varying intervals between measurements. Only 3 studies have used ABPM to estimate TTR. One study calculated 24-hour TTR in 228 patients with acute stroke,[11] another study used ABPM to estimate between-visit TTR,[7] and a third large individual-participant meta-analysis of 14 population-based cohorts used 24-hour ABPM with linear interpolation to quantify the time spent within systolic and diastolic BP (DBP) targets. This latter analysis demonstrated a strong inverse association between TTR and long-term mortality and adverse cardiovascular outcomes.[12]

This study assessed the association between TTR for SBP during 24-hour ABPM and both all-cause and cardiovascular mortalities in patients with sustained and masked hypertension (treated or untreated), controlled hypertension, and white-coat uncontrolled hypertension (both treated), based on office (≥140 and ≥90 mm Hg) and 24-hour BP measurements (≥130 and ≥80 mm Hg).

METHODS

Study Design

This study utilized data from the Spanish ABPM Registry, a nationwide observational cohort study conducted at 223 primary care centers within the Spanish National Health System. The baseline data, protocols, and inclusion-exclusion criteria have been previously reported.[9] A comprehensive database was created by requesting mortality information from the Spanish National Institute of Statistics and linking that data with the ABPM data. The data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Population

Between March 2004 and December 2014, the Spanish ABPM Registry included 51 942 adults aged ≥18 years with guideline-recommended indications for ABPM, including suspected white-coat hypertension, masked hypertension, high-risk hypertension, borderline hypertension, labile hypertension, and assessment of antihypertensive treatment efficacy. The study was sponsored by the Spanish Society of Hypertension, Lacer Laboratories, and European government agencies, which did not influence the study design, data analysis, or interpretation of results. Institutional review boards at each participating center approved the study protocols, and all patients provided written informed consent.

Because BP target ranges apply only to patients with an indication for antihypertensive therapy, the present analysis was restricted to patients fulfilling at least 1 of the following criteria:

1. sustained hypertension, defined as office BP ≥140 and ≥90 mm Hg and 24-hour BP ≥130 and ≥80 mm Hg, with or without antihypertensive medication;
2. masked hypertension defined as office BP <140/90 mm Hg but 24-hour BP ≥130 and ≥80 mm Hg, with or without antihypertensive medication;
3. white-coat hypertension in the presence of antihypertensive medication defined as elevated office BP (≥140 and ≥90 mm Hg) and controlled 24-hour BP (<130/80 mm Hg);
4. controlled hypertension in the presence of antihypertensive medication, defined as office BP of <140/90 mm Hg and 24-hour BP of <130/80 mm Hg.

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Procedures and BP Indices

Patient data were collected through standardized interviews and physical examinations during clinical visits. Office BP and heart rate were measured following established protocols. In 85% of patients, validated oscillometric devices were used, while 15% were measured using calibrated mercury sphygmomanometers. BP readings were taken, while the patients were seated and at rest. The mean of 2 BP readings was recorded.

ABPM was conducted using validated, automated devices (Spacelabs model 90207; Spacelabs Healthcare, Snoqualmie, WA) with appropriate cuff sizes, recording BP every 20 minutes during the day and every 30 minutes at night. The same criteria as in previous analyses of the Spanish ABPM Registry were applied: at least 70% of SBP and DBP successful readings during the daytime and nighttime periods, 24-hour duration, and at least 1 BP measurement per hour.[9,13] Mean 24-hour, daytime, and nighttime SBP and DBP were examined. Day and night periods were based on the patients' self-reported sleeping and waking times. TTR was calculated using linear interpolation, which was performed on a per-minute basis between 2 consecutive SBP readings obtained from ABPM and determined the proportion of time spent in the target range defined as an SBP of 120 to 134 mm Hg during daytime and an SBP of 110 to 119 mm Hg during nighttime (Figure S1). TTR was calculated as the number of SBP values (observed and interpolated) within the target range divided by the total number of SBP values (observed and interpolated), multiplied by 100. Interpolation was performed only when the interval between 2 consecutive readings was <60 minutes. In the absence of universally accepted ambulatory and office treatment targets, the SBP target range used for this analysis was defined as 120 to 134 mm Hg during daytime and 110 to 119 mm Hg during nighttime. This ambulatory BP range approximates an office SBP of 120 to 139 mm Hg and aligns with the recommended office BP targets, which the 2024 European Society of Hypertension guidelines recommend as a primary target for most patients.[14] The threshold of 120 mm Hg was chosen because the European Society of Hypertension guidelines advise against reducing office systolic BP <120 mm Hg, and the 2024 European Society of Cardiology guidelines define 120 mm Hg as the lower bound of the recommended office BP range for most treated patients.[14,15] For BP variability assessment, we chose 24-hour SBP coefficient of variation, calculated as the ratio between the SD and the mean of the SBP (×100).[16]

Mortality Data

The primary outcomes were all-cause and cardiovascular mortalities. Causes of death were determined from certificates coded according to the International Statistical Classification of Diseases. Follow-up began at the first recruitment visit and continued until the date of death or the end of the recruitment period on December 31, 2019. The Spanish National Institute of Statistics recorded the death data and verified its accuracy, completeness, and reliability.

Statistical Methods

Metric data were presented as means (SDs) and categorical data as frequencies (percentages). Differences between groups were tested using ANOVA for metric data and the χ^2 test for categorical data. Cox regression analyses were adjusted for age, sex, body mass index, number of antihypertensive medications (0 or untreated versus 1 versus 2 or more), previous cardiovascular disease (previous record of ischemic heart disease, stroke, or heart failure versus no record), diabetes (previous record of diabetes versus no record), dyslipidemia (previous record of dyslipidemia versus no record), and tobacco smoking status (current versus not) as recently reported.[9] Subgroup analyses were run for groups of mean 24-hour SBP, the same period as TTR, BP phenotype, and treatment (no antihypertensive medications versus ≥1 antihypertensive medications). There was no adjustment for the number of antihypertensive medications in subgroup analyses with BP phenotype and treatment because this variable would have limited the variance within the subgroups as the number of antihypertensive medications was 0 for all patients in the no treatment group, whereas, due to the exclusion criteria, included patients with normotension and white-coat hypertension were always on antihypertensive medications. The interaction between TTR and subgroup was tested with the likelihood-ratio test by adding the interaction to a model containing the adjustment variables, TTR, and the subgroup. If a significant interaction was found, the data were further divided by subgroups, and *P* values for TTR were determined using the likelihood-ratio test by adding TTR to a model containing the adjustment variables. This procedure applies to the results shown in Figures 2 and 3. In Table 2, hazard ratios (HRs) and CIs for TTR are obtained from Cox regression when analyses were additionally adjusted for mean 24-hour SBP and SBP variability. All Cox regression analyses were done for all-cause death and cardiovascular death. The trend of TTR over the lifespan was modeled with natural cubic splines using age terciles as knots and, in the case of trend by sex, including an interaction. *P* values are 2-sided with a significance level of 0.05 and remain unadjusted, that is, no correction for multiple testing was applied due to the exploratory nature of the analysis. All analyses were done with R, version 4.4.1.

RESULTS

Patient Characteristics

A total of 59 124 patients (47.0% female) were recruited.[9] After excluding 10 437 (17.7%) patients with a 24-hour SBP <130 mm Hg who were not receiving antihypertensive medication, 48 687 (82.3%) patients remained for the analyses. Among these, 45.6% were women. The mean age at baseline was 59.7 (SD, 13.6) years. The mean 24-hour SBP and DBP were 131.0 (SD, 13.7) mm Hg and 77.3 (SD, 10.5) mm Hg, respectively. During a median follow-up of 9.7 years, 6502 deaths occurred, including 2185 from cardiovascular causes. Table 1 shows the baseline characteristics of the population used for this analysis. Tables S1 through S3 summarize the baseline characteristics stratified by quartiles of 24-hour TTR, TTR <50% versus ≥50%, and hypertension phenotypes.

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Table 1. Baseline Characteristics

Variable	Statistic
No. of patients	48 687
Female, n (%)	22 215 (45.6)
Age, y	59.7 (13.6)
Body mass index, kg/m^2	29.1 (4.8)
Tobacco smoker, n (%)	7591 (15.6)
Diabetes, n (%)	10 014 (20.6%)
Dyslipidemia, n (%)	21 278 (43.7%)
History of disease	
CV disease, n (%)	6961 (14.3%)
Left ventricular hypertrophy, n (%)	2840 (5.8%)
Heart failure, n (%)	1029 (2.1%)
Stroke, n (%)	1972 (4.1%)
Chronic kidney disease, n (%)	984 (2.0%)
Antihypertensive treatment	
Treatment status	
None, n (%)	13 559 (27.8%)
Monotherapy, n (%)	12 543 (25.8%)
Combination therapy, n (%)	22 585 (46.4%)
Number of antihypertensive medications prescribed*	1.5 (1.3)
Diuretic, n (%)	17 282 (35.5%)
DHP calcium channel blocker, n (%)	10 029 (20.6%)
Non-DHP calcium channel blocker, n (%)	1346 (2.8%)
ACE inhibitor, n (%)	11 274 (23.2%)
Angiotensin receptor blocker, n (%)	18 945 (38.9%)
β-Blocker, n (%)	8736 (17.9%)
Alpha blocker, n (%)	2656 (5.5%)
Office BP	
SBP, mm Hg	149.5 (18.9)
DBP, mm Hg	87.0 (11.7)
Pulse rate, bpm	74.4 (14.2)
Ambulatory BP, mm Hg	
24-h SBP	5515 (11.3%)
24-h SBP <115, n (%)	17 541 (36.0%)
24-h SBP, 115–129, n (%)	25 631 (52.6%)
24-h SBP ≥130, n (%)	134.0 (14.1)
Daytime SBP, mm Hg	7378 (15.2%)
Daytime SBP <120, n (%)	18 827 (38.7%)
Daytime SBP, 120–134, n (%)	22 482 (46.2%)
Daytime SBP ≥135, n (%)	122.3 (15.8)
Nighttime SBP, mm Hg	10 487 (21.5%)
Nighttime SBP <110, n (%)	12 777 (26.2%)
Nighttime SBP, 110–119, n (%)	25 423 (52.2%)
Nighttime SBP ≥120, n (%)	5515 (11.3%)
24-h DBP, mm Hg	77.3 (10.5)
Daytime DBP, mm Hg	80.1 (11.1)
Nighttime DBP, mm Hg	69.3 (10.4)
24-h pulse rate, bpm	71.9 (10.5)
Daytime pulse rate, bpm	74.6 (11.4)
Nighttime pulse rate, bpm	65.6 (9.8)

Data are presented as n (%) or mean (SDs). ACE indicates angiotensin-converting enzyme; BP, blood pressure; CV, cardiovascular; DBP, diastolic blood pressure; DHP, dihydropyridine; and SBP, systolic blood pressure.

*Number of antihypertensive medications prescribed represents the mean (SDs) count of antihypertensive medications per patient and does not account for dose.

Time in Target Range

The mean 24-hour TTR was 29.3% (SD, 16.6%), ranging from 0% to 83.8%. During daytime (target range defined as SBP of 120–134 mm Hg), mean TTR was 32.7% (SD, 19.7%), while nighttime TTR (target range defined as SBP of 110–119 mm Hg) was 22.8% (SD, 19.4%). The 24-hour TTR decreased with age in both men and women (Figure 1; Figure S2).

Association of TTR With All-Cause Death

Higher 24-hour TTR was associated with a lower risk of all-cause death (HR, 0.83 per 1-SD increment [95% CI, 0.80–0.85]) in models adjusted for clinical confounders. This association remained significant after additional adjustment for mean 24-hour SBP (HR, 0.90 per 1-SD increment [95% CI, 0.87–0.93]) and SBP variability (HR, 0.91 per 1-SD increment [95% CI, 0.88–0.94]; Table 2). Significant interactions were observed between mean 24-hour SBP groups (≥130, 115–129, and <115 mm Hg), 24-hour TTR ($P_{interaction}$=0.0067; Figure 2), and nighttime TTR ($P_{interaction}$<0.0001) in relation to all-cause mortality, whereas the interaction with daytime TTR was not significant ($P_{interaction}$=0.3357; Figure S3). Specifically, higher TTR was associated with lower all-cause death among patients with elevated (≥130 mm Hg) and controlled (115–129 mm Hg) mean 24-hour SBP but not in those with low mean 24-hour SBP (<115 mm Hg; Figure 2). Similar interactions were observed for daytime and nighttime TTR with corresponding SBP levels (Figure S3). Higher TTR was associated with lower all-cause mortality in patients with elevated daytime and nighttime SBP, and in those with controlled daytime SBP but not in those with low SBP levels (Figure 2). Moreover, higher TTR was associated with higher all-cause mortality when stratifying for mean 24-hour DBP (≥80, 65–84, and <65 mm Hg), with the strongest association in the higher DBP groups (Figure S4). Across BP phenotypes, higher 24-hour TTR was associated with lower all-cause mortality in patients with masked hypertension, sustained hypertension (uncontrolled office and 24-hour BP, with or without antihypertensive medication), and controlled hypertension (controlled office and 24-hour BP while receiving antihypertensive medication) but not in those with white-coat uncontrolled hypertension receiving antihypertensive medication ($P_{interaction}$<0.0001; Figure 2). There was no significant interaction between treatment status (with versus without antihypertensive medication) and 24-hour, daytime, or nighttime TTR (Figure 2; Figure S3).

Association of TTR With Cardiovascular Death

Higher 24-hour TTR was also associated with a reduced risk of cardiovascular death (HR, 0.80 per 1-SD increment [95% CI, 0.76–0.84]). This association persisted

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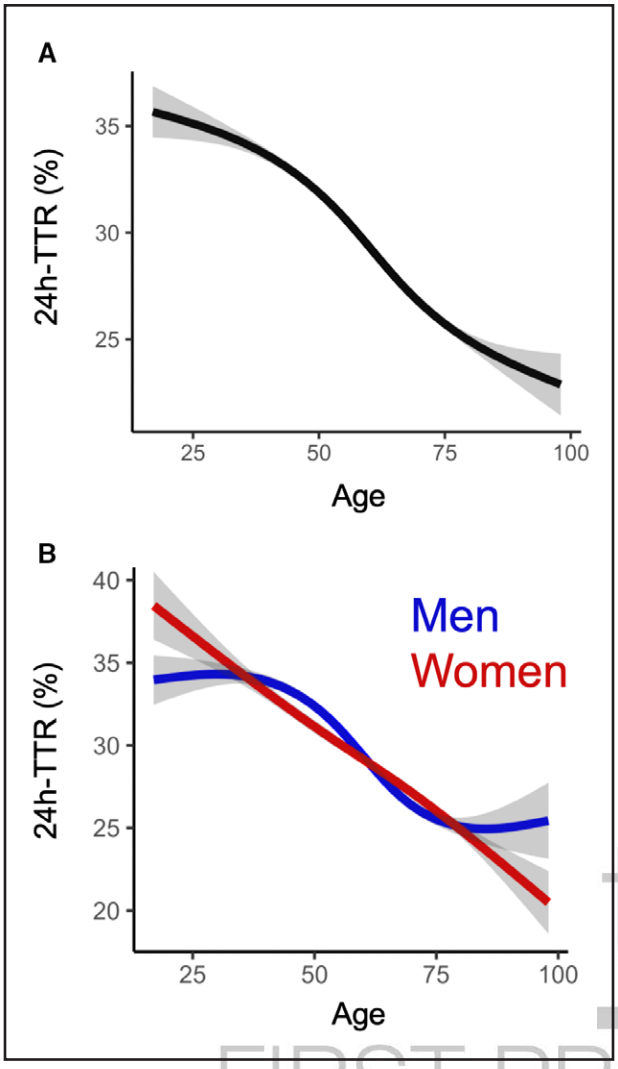

Figure 1. Time in target range (TTR) for systolic blood pressure (SBP) during 24-hour ambulatory blood pressure monitoring (ABPM) according to age and sex.
Associations between age and SBP TTR are measured by 24-hour ABPM for the overall cohort (**A**) and stratified by sex (**B**). Both men (blue) and women (red) exhibit age-related declines in TTR. Shaded areas represent 95% CIs.

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after adjustment for mean 24-hour SBP (HR, 0.91 per 1-SD increment [95% CI, 0.86–0.97]) and SBP variability (HR, 0.93 per 1-SD increment [95% CI, 0.88–0.99]; Table 2). A significant interaction was found between mean 24-hour SBP groups and nighttime TTR (but not 24-hour TTR or daytime TTR) for its association with cardiovascular death (Figure 3; Figure S5). As with all-cause mortality, high TTR was associated with reduced cardiovascular death in patients with elevated (≥130 mm Hg) and controlled (115–129 mm Hg) mean 24-hour SBP but not in those with low mean 24-hour SBP (<115 mm Hg; Figure 3). In addition, significant interactions were observed for daytime and nighttime TTR with corresponding SBP levels (Figure S5). Higher TTR during daytime and nighttime was associated with reduced risk of cardiovascular mortality only in patients with elevated SBP during the corresponding periods (Figure S5). There was no significant interaction between mean 24-hour DBP groups (≥80, 65–84, and <65 mm Hg) and 24-hour, daytime, or nighttime SBP TTR in relation to cardiovascular mortality (Figure S6). Across BP phenotypes, higher 24-hour TTR was associated with lower cardiovascular mortality in patients with masked hypertension and sustained hypertension (uncontrolled office and 24-hour BP, with or without antihypertensive medication) but not in those with white-coat uncontrolled hypertension receiving antihypertensive medication ($P_{interaction}$=0.0002) or those with controlled hypertension receiving antihypertensive medication (Figure 3). There was no significant interaction between treatment status (with versus without antihypertensive medication) and 24-hour, daytime, or nighttime TTR (Figure 3; Figure S5).

DISCUSSION

This analysis of 48 687 patients with hypertension from the Spanish ABPM Registry assessed the association between 24-hour TTR for SBP during ABPM and mortality outcomes. A higher 24-hour TTR was significantly associated with lower risks of both all-cause and cardiovascular mortalities, independent of mean 24-hour SBP and SBP variability. This association was observed in patients with and without antihypertensive medication. Stratified analyses revealed that the benefit of higher TTR was evident in patients with elevated SBP (≥130 mm Hg) and controlled SBP (115–129 mm Hg) but not in those with low SBP (<115 mm Hg). In addition, higher TTR was associated with lower mortality risk in patients with masked hypertension, sustained hypertension, and controlled hypertension in the presence of antihypertensive medication, whereas no such association was observed in patients with white-coat uncontrolled hypertension.

TTR reflects the proportion of time a patient remains within a predefined target range over multiple time points. Most prior studies assessing TTR have relied on infrequent office or home BP measurements, typically taken only a few times per year.[17] When TTR is calculated from sparse measurements using linear interpolation, it may fail to capture rapid or nonlinear fluctuations, hence oversimplifying the data. In addition, because the intervals vary, some interpolated segments may cover more time than others.[18] Therefore, calculating TTR based on only a few snapshot measurements can be more susceptible to outliers. These outliers may arise from intrinsic or extrinsic factors, such as stress, physical activity, dietary changes, seasonal variations, or improper BP measurement techniques, and can bias the estimates.[4,16] Moreover, when narrow target ranges are applied, TTR becomes inherently sensitive to both pathological and physiological BP variability.

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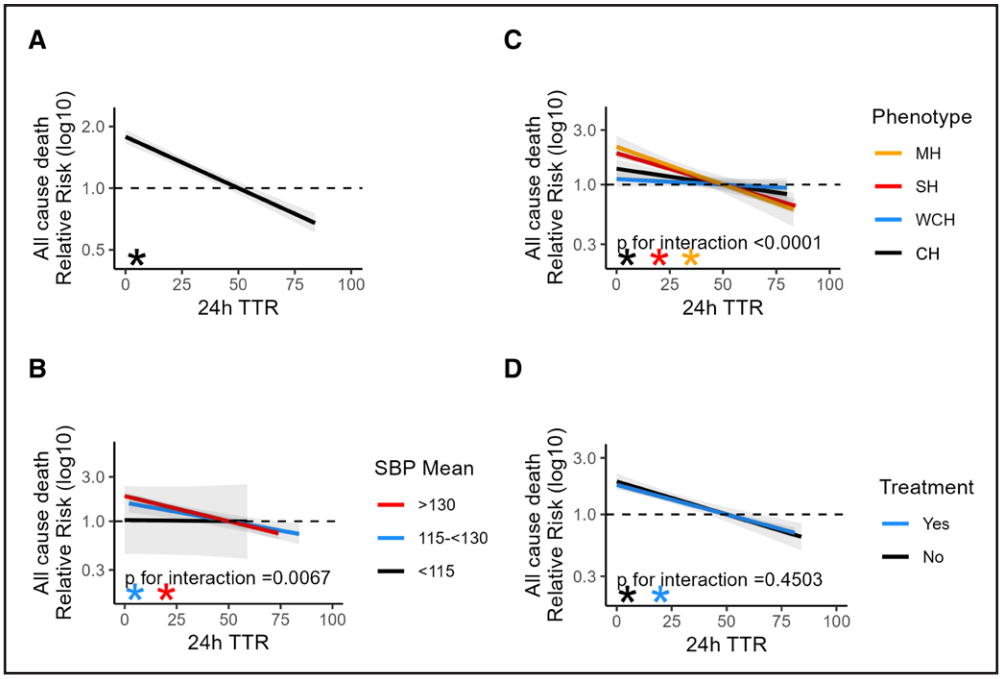

Figure 2. Association of time in target range (TTR) for systolic blood pressure (SBP) during ambulatory blood pressure monitoring and relative risk for all-cause death.

A, All patients. Cox regression analyses were used to examine the association between SBP TTR and all-cause mortality, adjusted for age, sex, body mass index, number of antihypertensive medications, history of cardiovascular disease, diabetes, dyslipidemia, and smoking status. Solid lines represent adjusted Cox model–estimated hazard ratios across increasing SBP TTR, and shaded areas indicate 95% CIs. Hazard ratios are expressed relative to the predicted hazard at a 50% TTR within each subgroup. Analyses were stratified by mean 24-hour SBP levels (**B**), hypertension phenotype (**C**), and treatment status (**D**). Interaction between TTR and subgroup was assessed using likelihood-ratio tests by adding an interaction term to a model containing the adjustment variables, TTR, and the subgroup. When a significant interaction was identified, subgroup-specific associations were evaluated, and *P* values for TTR were derived by adding TTR to a model containing the adjustment variables. Asterisks denote statistically significant associations ($P<0.05$) between TTR and all-cause mortality in subgroups with a significant interaction. CH indicates controlled hypertension; MH, masked hypertension; SH, sustained hypertension; and WCH, white-coat hypertension.

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Observational studies from large databases, including the Veterans Affairs system[19] and primary care records in England,[8] showed an inverse association between TTR (based on office SBP) and all-cause mortality at 10 years and incident cardiovascular disease through 5 years. Similarly, a post hoc analysis of the SPRINT trial demonstrated that greater TTR, based on office SBP over 3 months, was associated with reduced risk of major cardiovascular events, even after adjustment for mean SBP and SBP variability.[6] In contrast to most previous studies that examined the relationship between TTR and cardiovascular outcomes based on a limited number of office BP measurements, this study derived TTR from 24-hour ABPM, providing a more comprehensive and physiologically relevant assessment of BP exposure. Before this study, only 3 studies have used ABPM to estimate TTR. One study calculated 24-hour TTR in 228 patients with acute stroke,[11] another study used ABPM to estimate between-visit TTR,[7] and a third individual-participant meta-analysis of 14 population-based cohorts used 24-hour ABPM with linear interpolation to quantify the time spent within SBP and DBP targets. This latter meta-analysis demonstrated a strong inverse association between TTR and long-term mortality and adverse cardiovascular outcomes.[12]

This study extends these findings in several important ways. It included nearly 3 times as many participants as the meta-analysis (48 687 versus 14 230) and was based on a contemporary cohort recruited from 223 primary care centers. Moreover, higher TTR was associated with a lower risk of all-cause and cardiovascular mortalities even after adjusting for mean 24-hour SBP and SBP variability. Notably, when stratifying by SBP categories, the association between TTR and mortality was not only significant in patients with elevated SBP but also in those with controlled SBP. This highlights the immediate and clinically relevant added value of 24-hour TTR compared with mean 24-hour SBP, suggesting its potential clinical relevance in guiding hypertension management. Importantly, the association between TTR and mortality remained significant across all DBP subgroups.

Although this analysis demonstrated a linear inverse association between 24-hour TTR and mortality, whether this relationship extends to adverse events remains uncertain. Notably, the SPRINT post hoc analysis found no association between TTR (based on office

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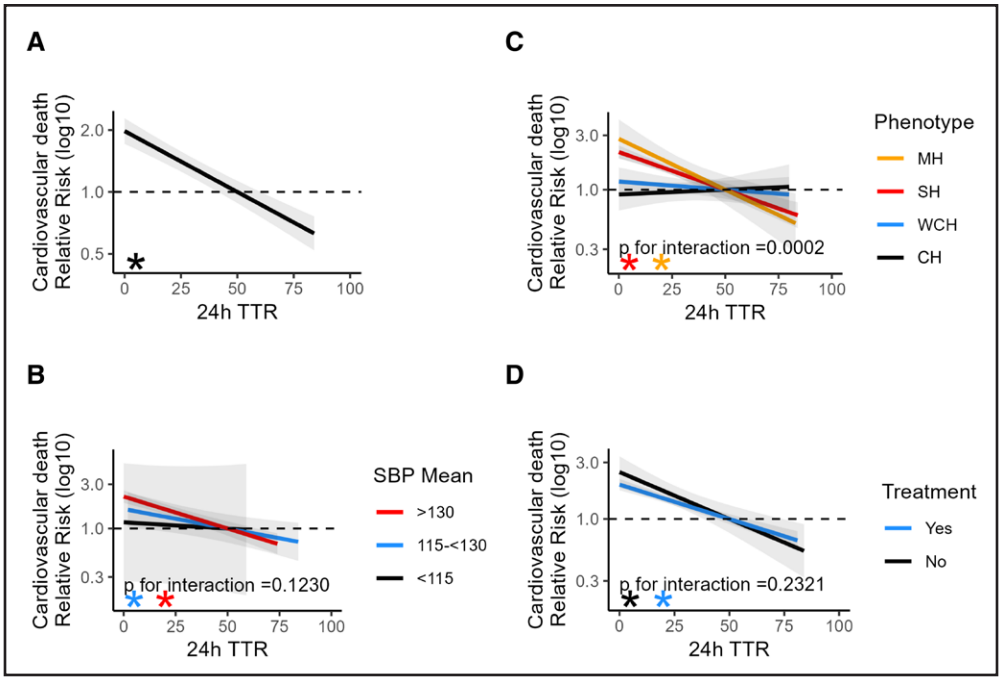

Figure 3. Association of time in target range (TTR) for systolic blood pressure (SBP) during ambulatory blood pressure monitoring and relative risk for cardiovascular (CV) death.

A, All patients. Cox regression analyses were used to examine the association between SBP TTR and CV mortality, adjusted for age, sex, body mass index, number of antihypertensive medications, history of cardiovascular disease, diabetes, dyslipidemia, and smoking status. Solid lines represent adjusted Cox model–estimated hazard ratios across increasing SBP TTR, and shaded areas indicate 95% CIs. Hazard ratios are expressed relative to the predicted hazard at a 50% TTR within each subgroup. Analyses were stratified by mean 24-hour SBP levels (**B**), hypertension phenotype (**C**), and treatment status (**D**). Interaction between TTR and subgroup was assessed using likelihood-ratio tests by adding an interaction term to a model containing the adjustment variables, TTR, and the subgroup. When a significant interaction was identified, subgroup-specific associations were evaluated, and *P* values for TTR were derived by adding TTR to a model containing the adjustment variables. Asterisks denote statistically significant associations ($P<0.05$) between TTR and CV mortality in subgroups with a significant interaction. CH indicates controlled hypertension; MH, masked hypertension; SH, sustained hypertension; and WCH, white-coat hypertension.

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SBP) and serious adverse events, including syncope and injurious falls.[6] In addition, pooled data from the ACCORD-BP trial (Action to Control Cardiovascular Risk in Diabetes-Blood Pressure) and SPRINT showed that higher office-based TTR was linked to lower risks of major cardiovascular and adverse kidney events.[20] In contrast to the association between TTR and cardiovascular events, the relationship with renal events was nonlinear, suggesting that modest increases in TTR may be sufficient to reduce risk, while further risk reduction only occurred at high TTR levels.[20] This has been attributed to renal autoregulation.[20] However, the potential influence of age, sex, comorbidities, and BP phenotypes (eg, orthostatic hypotension or autonomic dysfunction) on the relationship between TTR and clinical outcomes remains to be fully elucidated. In such populations, mean 24-hour SBP may seem adequate, but significant BP variability, which is partly captured by 24-hour TTR, could be associated with symptoms and adverse clinical events.[21]

Table 2. Association of Time in Target Range for 24-h SBP and Risk for All-Cause and CV Deaths

	Model 1*	Model 2 (additionally adjusted for mean 24-h SBP)	Model 3 (additionally adjusted for mean 24-h SBP and SBP variability)
	HR per 1-SD increment (95% CI)	HR per 1-SD increment (95% CI)	HR per 1-SD increment (95% CI)
All-cause death	0.83 (0.80–0.85)	0.90 (0.87–0.93)	0.91 (0.88–0.94)
CV death	0.80 (0.76–0.84)	0.91 (0.86–0.97)	0.93 (0.88–0.99)

Values are HRs per 1-SD increment with 95% CIs. CV indicates cardiovascular; HR, hazard ratio; and SBP, systolic blood pressure.

*Model 1 was adjusted for age, sex, body mass index, number of antihypertensive medications, previous cardiovascular disease, diabetes, dyslipidemia, and tobacco smoking status.

Limitations

Some limitations of this study warrant discussion. First, this analysis focused on SBP, given its superior prognostic value and current guideline emphasis on SBP as the primary treatment target.[14] DBP often declines with advancing age due to arterial stiffening. Second, although ABPM provides more frequent measurements than office BP, the number of readings over 24 hours may still be insufficient to characterize TTR and BP variability in some patients fully.[22] Third, ABPM was

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performed only once at baseline, and acute or transient influences during monitoring could affect the results. Fourth, information on antihypertensive treatment was not captured during follow-up, limiting the ability to adjust for treatment changes over time. Fifth, data on symptomatic hypotension at baseline and on potential complications related to excessive BP lowering, such as symptomatic hypotension, syncope, falls, or acute kidney injury, were not captured in the Spanish ABPM Registry and, therefore, could not be assessed during follow-up. Sixth, the cause of death classification relied on death certificates, which may underestimate cardiovascular mortality due to misclassification. Seventh, the study cohort consisted exclusively of White individuals from Spain, and thus, it may not apply to other ethnicities or regions.

Perspectives

In patients with treated or untreated sustained or masked hypertension, white-coat uncontrolled hypertension, and controlled hypertension, a higher TTR for SBP, as measured by ABPM, was independently associated with lower risks of both all-cause and cardiovascular mortalities. These associations remained significant even after adjustment for mean 24-hour SBP and SBP variability. These findings underscore the potential value of 24-hour TTR as a complementary metric of BP level and variability for risk stratification in hypertension. Future studies are warranted to determine whether interventions specifically aimed at increasing 24-hour TTR translate into improved clinical outcomes and to explore the role of TTR in guiding treatment strategies.

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Supplemental Material

Tables S1–S3
Figures S1–S6

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